The World Did Not Come with a Formula: How Our Representations Shape What We Can Know

When I was in high school, I began asking a question that seemed simple but was surprisingly difficult to answer: Why were we learning all these things? I was not only questioning whether a particular subject would be useful in my future. I was questioning why human knowledge had taken the form it had. Why did mathematics look the way it did? Why were we expected to understand the world through these particular formulas, concepts, and categories? Even when a theorem could be proven, I wondered why the underlying truth had to appear in that particular representation. Surely, the truth itself did not come into existence when someone discovered a way to express it. It had been there all along.

Eventually, this question expanded beyond mathematics. Why did human civilization develop the way it did? What if the earliest civilizations had organized knowledge differently? Would we still have developed the same scientific theories, educational institutions, or social structures? I even wondered whether an entirely different civilization could have emerged from the same physical world. At the time, these questions felt too abstract to investigate. Studying cognitive science has given me a different way of approaching them. Perhaps the question is not simply why civilization developed as it did, but how the human mind's ways of representing information make certain developments possible while making others more difficult to imagine.

My central argument is that representations do more than describe reality: they influence the computations we can perform, the relationships we notice, and the questions we consider worth asking. Because new knowledge is built using representations inherited from earlier generations, this creates a process I call *representational inheritance*. Human knowledge is constrained by the world it attempts to explain, but its development is also shaped by the cognitive tools through which that world becomes understandable.

An unexpected example of this possibility comes from the *I Ching*, or *Book of Changes*, an ancient Chinese text that began as a divination tradition and later became an important source of philosophical reflection. One of its most interesting features is its use of two basic kinds of lines, broken and unbroken, arranged into six-line figures called hexagrams. These combinations produce sixty-four distinct patterns, each associated with traditional interpretations of situations and change. The system does not claim that reality literally consists of sixty-four physical states. Rather, it provides a symbolic framework through which an immense variety of human situations can be interpreted.

What interests me is not whether the *I Ching* can predict the future. I am interested in the cognitive problem that its existence reveals. Human experience is continuous and extraordinarily complicated, yet the *I Ching* approaches this complexity through a limited collection of symbols and relationships. Events can be considered in terms of movement, opposition, timing, and transformation. Instead of treating a situation as a collection of isolated facts, this framework encourages a person to consider how those facts might belong to a larger process. The representation makes some relationships especially salient.

But suppose an ancient civilization had developed a completely different system for interpreting change. Perhaps it would have emphasized numerical probabilities, spatial structures, or causal networks rather than symbolic transformations. People using these systems might have attended to different aspects of an otherwise identical situation. They would not necessarily have inhabited different physical worlds. They might, however, have organized their experiences into different cognitive worlds.

This distinction becomes clearer through Marr's three levels of analysis. Marr proposed that understanding an information-processing system requires distinguishing the computational problem being solved, the representations and algorithms used to solve it, and the physical mechanisms implementing those processes (Marr, 1982). These levels help prevent us from confusing a representation with the reality it represents.

Consider a person using the *I Ching* to think through a difficult decision. At the computational level, we might ask what problem the person is trying to solve. Perhaps they are trying to reason about an uncertain situation and decide how to act. At the algorithmic and representational level, we can examine how symbols, distinctions, and interpretive procedures organize their thinking. At the implementation level, we would investigate the actual cognitive and neural mechanisms involved in reading, remembering, and interpreting those symbols.

This does not mean that the *I Ching* is itself a complete cognitive algorithm, or that its interpretations are scientifically validated. Its interpretive rules are not equivalent to the precisely specified transitions of a computer program. Rather, it demonstrates how a culturally developed symbolic system can become a tool through which a person organizes a problem. Marr's framework helps us ask how such a tool might support thinking without mistaking its philosophical claims for an explanation of how the brain works.

This connection also reminds me of our discussion of Turing machines. A Turing machine processes symbols according to rules that specify how one state transitions into another. What makes the model important for cognitive science is that complex information processing can arise from operations over comparatively simple representations. Symbols can be combined, transformed, and reused, allowing a limited system to generate many possible expressions and computations.

Yet knowing that a system can perform a computation is different from knowing how efficiently it can do so. A representation may contain all the information necessary to solve a problem while making the solution extremely difficult to discover.

Larkin and Simon (1987) examined why diagrams can sometimes make problem-solving more efficient than verbal descriptions containing equivalent information. In a diagram, relationships that would otherwise need to be inferred may become spatially explicit. The person solving the problem can take advantage of how information is arranged instead of mentally reconstructing every relationship. Similarly, Zhang and Norman (1994) investigated how internal and external representations jointly contribute to cognitive tasks. Their work suggests that the structure of a representation can change how a task is performed, even when its underlying abstract problem remains the same.

This provides a possible answer to my childhood question about mathematical formulas. Suppose the same mathematical relationship can be expressed through a geometric construction or an algebraic equation. Both may be correct. But they do not necessarily invite the same mental operations. A diagram may make a spatial relationship immediately apparent, while an equation may make symbolic manipulation easier. Neither changes the mathematical relationship itself. What changes is how accessible particular inferences become.

The same principle becomes even more interesting when extended beyond individual problem-solving. If one representation makes a particular inference easier, people may discover certain relationships sooner. Those discoveries may motivate new questions, which then lead to additional representations. Over time, the intellectual possibilities of later generations become partly dependent on the tools developed by earlier generations.

This is what I mean by representational inheritance.

I imagine the process beginning with a basic problem. A generation develops a representation to make that problem manageable. Through repeated use, the representation becomes familiar and useful. It is eventually taught to others, who begin using it not only to solve the original problem but also to investigate new ones. These later discoveries may produce increasingly sophisticated representations, which become the foundations for further developments.

In this sense, human knowledge is recursive: the output of one period of intellectual activity becomes part of the input for the next. However, the process is not literally a recursive computer program governed by one fixed rule. The analogy concerns how previously developed cognitive tools become resources and constraints for later thinking.

Consider the development of mathematical notation. Once a symbolic system makes particular operations convenient, people can use it to discover and express relationships that would have been more difficult to manipulate using older methods. Those discoveries are incorporated into education, scientific research, and technological development. Future researchers inherit a world already organized around them. They gain access to remarkable intellectual resources, but they may also become accustomed to asking questions in terms established by those resources.

An important consequence follows: **representations may influence not only how we answer questions, but which questions occur to us in the first place.**

This idea connects to the frame problem discussed in cognitive science. When facing a complex situation, how does a cognitive system determine which information is relevant? Fodor (1983) characterized central cognition as involving potentially wide-ranging relationships among beliefs. In principle, almost any piece of background knowledge might become relevant to a particular judgment. Yet a mind with limited time and resources cannot examine every possible connection.

Representational systems offer one way to manage this difficulty. By organizing information into categories and relationships, they make some possibilities easier to consider and others less

immediately accessible. A representation helps reduce the space of possibilities that must be searched.

But that convenience creates a trade-off. When a system highlights certain variables, it can make us less likely to consider variables outside its structure. For example, an economic model might encourage us to evaluate a decision through costs, incentives, and expected outcomes. A moral framework might encourage us to consider obligations and responsibilities. Neither framework necessarily contains false information. They organize the relevant information differently.

This reminds me of my own education. I have studied statistics, economics, and computer science, and each discipline provides powerful tools for understanding problems. Yet knowledge is often taught as though it belongs to separate compartments. It is only when trying to understand a real situation that I begin to see how these areas might connect. I wonder whether some of the difficulty comes from the way knowledge is organized and taught. We learn to become competent within particular representations, but receive comparatively little instruction in how to move between them.

If representational inheritance influences individual learning, could something similar shape civilization?

I think it could, although this claim requires caution. Historical development is influenced by physical environments, politics, social institutions, material resources, and countless other factors. Cognitive representation cannot explain civilization by itself. Nevertheless, the concepts available within a culture may help determine which problems can be formulated, communicated, and investigated. Once certain ways of thinking become established in educational and institutional practices, they may become easier to preserve and extend than unfamiliar alternatives.

This suggests that the development of knowledge could be path-dependent. Even if two civilizations confronted similar physical problems, differences in their early representational tools might influence the sequence of discoveries that followed. One might organize a domain primarily through quantitative relationships, while another might initially emphasize classifications or qualitative transformations. Their later developments would be shaped by the intellectual resources already available.

Still, I do not think this means that knowledge is arbitrary.

Here, Marr's discussion of constraints from above and below offers an important correction to my argument. At the computational level, the problem a system must solve and the regularities of the environment constrain what kinds of solutions are appropriate. At the implementation level, physical mechanisms constrain what processes can actually be realized. A representational system cannot simply ignore these requirements and remain successful.

The same principle applies to human knowledge. Different mathematical notations can express the same valid relationship, but the relationship does not become true merely because someone invents a notation for it. Engineering methods must still respect physical properties if the

structures they produce are to function. Scientific theories must ultimately confront evidence. Reality places constraints on which of our representations can succeed.

I therefore think two claims can coexist. First, there may be many possible ways to represent and investigate the same world. Second, these representations are not equally effective for every task, and some will fail when tested against reality.

The fact that multiple paths are possible does not mean every destination is possible.

A serious objection to representational inheritance is that, given enough time, rational investigators would eventually discover the same truths regardless of their starting representations. Perhaps different civilizations would develop different symbols but ultimately converge on equivalent scientific explanations.

I find this objection persuasive in one respect. It would be unreasonable to assume that physical laws depend on the culture investigating them. However, convergence on certain truths does not require convergence on every method, concept, or question. Two representations can be logically compatible with the same facts while differing greatly in computational efficiency. The limitations of attention, memory, and time may make some discoveries accessible within one framework but prohibitively difficult within another.

More importantly, even if two societies eventually reach equivalent conclusions about a particular problem, they may arrive there through different sequences of intermediate discoveries. Those intermediate steps can create technologies, institutions, and new intellectual problems that would not otherwise have appeared in the same form. A common reality may constrain the range of successful knowledge systems without uniquely determining the history of their development.

There is another distinction I need to make. Saying that representations influence cognition does not necessarily mean that beliefs alter basic perception. In our discussion of cognitive penetrability, we considered the difficulty of demonstrating that higher-level knowledge directly changes perceptual processing rather than attention, interpretation, or judgment. Firestone and Scholl (2016) argue that many apparent effects of cognition on perception can be explained without establishing genuine cognitive penetration.

My argument is narrower. I am not claiming that someone who learns a new mathematical notation literally sees a different physical world. I am proposing that their representation changes how they organize information, perform inferences, and identify meaningful questions. These effects could occur even if the perceptual processes supplying their initial information remained unchanged.

In fact, this distinction makes the hypothesis more interesting because it suggests a way to test it.

Imagine an experiment in which participants receive exactly the same information about a fictional city undergoing economic, environmental, and social changes. One group receives the information in a table organized by separate categories. Another group receives a diagram

emphasizing relationships and changes over time. After studying the information, both groups answer questions about the city and are asked to propose new questions that future researchers should investigate.

If my hypothesis is correct, the two groups may differ not only in their ability to draw particular inferences but also in the questions they generate. Participants using a relational diagram might be more likely to investigate connections and feedback effects, while those using separate categories might focus more heavily on comparisons within individual domains. This is a prediction, not an established finding, and the design would need to control carefully for differences in how information is presented.

The experiment could then introduce a second generation of participants. Instead of receiving the original questions, they would inherit questions generated by the first group. We could examine whether differences in the initial representation continue to influence what the next group chooses to investigate, even when everyone has access to the same underlying facts.

A laboratory experiment could not establish how entire civilizations develop. But it could test a smaller cognitive mechanism underlying the argument: whether a representation influences question generation, and whether those questions subsequently constrain another person's reasoning. Protocol analysis, patterns of error, and processing times could help reveal how participants arrive at their conclusions.

This possibility changes how I think about the cognitive approach itself. We often begin by identifying a behavior and attempting to reconstruct the information-processing steps that produce it. But perhaps we should sometimes move one step further back and ask why the problem was represented in that particular way. A cognitive system might perform its computations perfectly while operating within a representation that makes certain possibilities difficult to express.

Returning to the *I Ching*, I no longer see its relevance to cognitive science primarily in its claims about the nature of change. I see it as one historical example of how people create symbolic tools to make an overwhelming world intellectually manageable. Its significance for my argument lies not in whether its interpretations are correct, but in what its structure reveals about the human attempt to transform experience into something that can be considered, discussed, and acted upon.

This also brings me back to the question I first asked in high school. I was frustrated because knowledge seemed to arrive already organized, as though its existing form were inevitable. Now I recognize that the organization itself is part of what deserves investigation. The formulas, categories, and theories we inherit are extraordinary intellectual achievements. But they are also products of earlier attempts to make sense of reality.

I do not believe human civilization is simply trapped inside its own representations. If that were true, we could never revise our theories, invent new methods, or discover that our previous assumptions were mistaken. The world continually challenges our descriptions of it, and humans are capable of creating new representations in response.

Yet our ability to change a representation does not mean we are always aware of the assumptions embedded within it. Perhaps one of the most valuable functions of cognitive science is to make those assumptions visible.

I began with the question of whether civilization could have developed differently. I cannot answer that historical counterfactual with certainty. But I now think the more productive question is whether our inherited representations influence which intellectual paths we are most likely to explore.

Perhaps intellectual progress requires two different achievements: discovering new answers within our existing representations, and occasionally recognizing that a different representation would allow us to ask a question we had never thought to ask.

The world may have truths waiting to be discovered. But it did not come with instructions telling us which symbols to use, which questions to prioritize, or which paths of discovery to follow.

Those are partly our own constructions. And understanding how we construct them may be just as important as understanding what they allow us to discover.

References

Firestone, C., & Scholl, B. J. (2016). Cognition does not affect perception: Evaluating the evidence for "top-down" effects. *Behavioral and Brain Sciences, 39*, e229. https://doi.org/10.1017/S0140525X15000965

Fodor, J. A. (1983). *The modularity of mind*. MIT Press.

Larkin, J. H., & Simon, H. A. (1987). Why a diagram is (sometimes) worth ten thousand words. *Cognitive Science, 11*(1), 65–100. https://doi.org/10.1111/j.1551-6708.1987.tb00863.x

Marr, D. (1982). *Vision: A computational investigation into the human representation and processing of visual information*. W. H. Freeman.

Wilhelm, R. (1967). *The I Ching or Book of Changes* (C. F. Baynes, Trans.; 3rd ed.). Princeton University Press.

Zhang, J., & Norman, D. A. (1994). Representations in distributed cognitive tasks. *Cognitive Science, 18*(1), 87–122. https://doi.org/10.1207/s15516709cog1801_3